

Consumer Guide: Safe Charging for Your Electric Vehicle

Ensuring Quality and Safety at Zimbabwe's New Charging Stations

As of March 2026

Abstract

As Harare and Bulawayo transition toward a greener commute, the new **2026 Energy Governance Framework** has established a "Floor of Quality." This guide helps you identify authorized charging points and avoid the risks associated with sub-standard electrical infrastructure.

1. The "Safety First" Checklist

Before plugging in your vehicle at any public or residential charging station, verify the following physical safety markers:

- **ZERA Certification Sticker:** Look for the 2026 holographic seal. This confirms the station meets the latest *Minimum Energy Performance Standards*.
- **Cable Integrity:** Ensure the cable is not frayed, kinked, or taped. Sub-standard "gray market" cables are the primary cause of thermal runaway.
- **IP54 Weatherproofing:** Public stations must be rated IP54+ for Zimbabwe's summer downpours. Avoid use during rain if the casing looks like standard indoor plastic.

1 Identifying the "Floor of Quality"

To protect your EV's high-value battery investment, ensure the station provides these mandatory technical protections:

A. Surge Protection (The "ZESA Guard")

Authorized stations must have integrated **Type 2 Surge Protection**. ***Risk:** Without this, a grid spike can "fry" your On-Board Charger (OBC), a repair costing thousands of dollars.*

B. Ground Fault Monitoring

A quality station performs a "Self-Test" upon plug-in.

- **Green Light:** "Ready" or "Ground OK."
- **Red Light:** "Fault" or "Earth." **Do not attempt to charge.**

C. Thermal Throttling

Standard-compliant chargers include internal heat sensors. If the station exceeds **60°C**, the system must automatically slow or stop the charge.

2 Charging Best Practices: Harare & Bulawayo

gray!15 Action	Why it Matters
Avoid "DIY" Adapters	Locally "modified" plugs can melt your home's wiring.
Check the RCD	Ensures the station is connected to a Residual Current Device—your "life insurance" against shorts.
Report Sub-standard Gear	If the ZERA seal is missing or gear is melting, report it to the Energy Governance Board.

4. EMERGENCY PROTOCOL

In the event of **smoke, sparks, or a "burning plastic" smell**:

1. **Press the Emergency Stop:** Use the large, red "E-Stop" button.
2. **Disconnect (If Safe):** Only disconnect if there is no visible flame.
3. **Alert the Operator:** Call the number on the ZERA certification sticker.

Protect your car. Protect your home. Stick to the Standard.